Real-World Order of Operations

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lessons 7–8 — Real-World Order of Operations	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not. No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes I–III, both Examples and boards 2, 3, 4, 6, 7 and 9 — the expression arrives inside a situation, and the unit is part of the answer.
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II and board 9 — the same three numbers are written as two different expressions, so the structure is seen to come from the situation and not from the page.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits over whether the flat fee goes inside the parentheses, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Essential question. If key words cannot tell you which operation a problem needs, what can?

Building on. 4.05–4.06 — grouping symbols and the four levels; MA.6.AR.1 — substituting a value for a letter; Unit 2 — rounding at the end, not in the middle.

Working toward. The Unit 4 assessment; MA.8.AR.2 and MA.912.NSO.1.4, where formulas are rearranged rather than only evaluated.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two problems have the same numbers and the same words and different answers. What is it in the problem, then, that decides the operation?	DOK 3
Why does rounding √200 to 14 before multiplying by 6 lose most of a foot, when 14 is only about 0.14 away from √200?	DOK 3
In a cost expression, what has to be true of a price for it to belong inside the parentheses that are multiplied by the area?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Evaluate $3x^2$ when $x = -2$.	***12***	−12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to 3x [8.NSO.1.7: the exponent belongs to its own base]
2	A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is K = ½mv², for a mass m and a speed v. Find her kinetic energy.	***2,240 joules***	anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as 2v [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules
3	Renting a kayak costs 17 dollars plus 9 dollars for each hour. Find the cost of 3 hours.	***44 dollars***	78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]; 29 — 17 + 9 + 3, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]
4	A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost.	***284 dollars***	4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — 8(3.50) + 60: the side used instead of the area
5	Evaluate ½mv² when m = 40 and v = 3. Which is the value?	***A — 180***	(B) v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]; (C) the one half was dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, $(½ \cdot 40 \cdot 3)^2$ [8.NSO.1.7: the exponent belongs to the v alone]
6	A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost?	***D — $12^2(1.25) + 38$***	(A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate]
7	A square garden has 4 sides, each √150 feet long. Find its perimeter, to the nearest hundredth of a foot.	***48.99 feet***	48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]; 48.99 with no unit — the answer is in feet
8	Evaluate $(a^2 - b) \div c$ when a = 7, b = 4, c = 5.	***9***	48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — 49 ÷ 5 done first, then 4 subtracted
9	One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings. Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery. Find both totals, and say why the key words could not have told you.	***23 dollars and 31 dollars***	23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 23 twice — the second structure was not read

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give a two-step frame — WRITE THE EXPRESSION, then EVALUATE IT — and require the expression on paper before any arithmetic. Substituted values go in parentheses every time, with the parentheses pre-printed on the frame for the first two problems.

ELL. Rate and flat fee are the two phrases every cost problem turns on, and 'a square foot' is what marks the difference. Put 'dollars a square foot' and 'a flat 60 dollars' side by side on the board and ask which one depends on the size of the floor, before any expression is written.

Enrichment. Ask for a third problem using 15, 4 and 6 whose answer is neither 39 nor 66. Then ask whether every arrangement of three numbers and two operations can be made into a sensible situation.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to give both totals and say what decides the operation, since the words and the numbers are identical in the two problems. Read the boards, not the papers.

Vocabulary. **formula**; **substitute**; **unit**

Materials. Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the unit test: Evaluate variable expressions: word problems — DPG.